

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

cf_d.sol_{ver} team (solver v0.2.0)

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Abstract

This report documents the benchmark results of `cfd.solver` v0.2.0, a 2-D unstructured cell-centered finite-volume solver for the compressible Euler and laminar Navier–Stokes equations, on the eight-case agentic CFD benchmark. The solver combines second-order least-squares reconstruction with a Venkatakrishnan limiter, the Rusanov approximate Riemann solver, laminar viscous fluxes, matrix-free LU-SGS pseudo-time marching for steady cases, and a true BDF2 outer physical-time loop with inner iterations for the transient cylinder case. Meshes are partitioned with METIS k -way, and halo exchange uses neighbor-scoped `isend/irecv` without replicating the mesh or the state on any rank.

Six of the eight cases converged to their target residual reduction on 4 MPI ranks: all five NACA0012 cases except the $M_\infty = 0.8$ inviscid case, and the cylinder $\text{Re} = 20$ case. The $M_\infty = 0.8$ inviscid run *failed to converge*: it stalled near one order of residual reduction with repeated positivity fallback (CFL halving) and was terminated without producing final field or surface output. The cylinder $\text{Re} = 200$ transient case was *not run* to completion and produced no results; only a 20-step smoke test (physical time $t = 0.2$) exists, confirming that the BDF2 machinery executes and converges its inner iterations.

Key converged results: NACA0012 at $M_\infty = 0.15$ inviscid gives $C_D = 0.0183$ and $C_L \approx -2.3 \times 10^{-4}$ (zero-lift symmetry satisfied to $< 1 \times 10^{-3}$); the laminar $\text{Re} = 5000$ airfoil cases give $C_D = 0.239$ ($M_\infty = 0.15$), 0.122 ($M_\infty = 0.8$) and 0.106 ($M_\infty = 2.0$); the cylinder $\text{Re} = 20$ case gives $C_D = 2.812$ with a steady symmetric wake. All final fields satisfy positivity of density and pressure, no-slip walls show exactly zero velocity, and slip walls show zero normal velocity to machine precision. Known limitations are documented in the final section.

1 Introduction

The benchmark objective is to validate a 2-D compressible CFD solver on a fixed set of eight cases covering subsonic, transonic, and supersonic inviscid and laminar flow over a NACA0012 airfoil, plus steady and time-periodic laminar flow over a circular cylinder. The required cases are:

- NACA0012, $M_\infty = 0.15$ and $M_\infty = 0.8$ and $M_\infty = 2.0$, inviscid (slip wall);
- NACA0012, $M_\infty = 0.15$ and 0.8 and 2.0 , laminar $\text{Re} = 5000$ (no-slip adiabatic wall);
- circular cylinder, $M_\infty = 0.1$, laminar $\text{Re} = 20$ (steady);
- circular cylinder, $M_\infty = 0.1$, laminar $\text{Re} = 200$ (unsteady vortex shedding, transient BDF2).

The report is written as a technical CFD report, not a run log: numerical choices are connected to observed results, and the two incomplete cases ($M_\infty = 0.8$ inviscid and cylinder $\text{Re} = 200$) are reported honestly as failures/pending rather than as successes.

2 Governing Equations and Nondimensionalization

The conservative state vector in two dimensions is

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T,$$

and the 2-D compressible Navier–Stokes equations in conservative form read

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}.$$

The inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix},$$

with the calorically perfect gas closure

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho},$$

using $\gamma = 1.4$ and $R = 1$ throughout. The viscous fluxes follow from the Newtonian stress tensor

$$\tau_{xx} = \frac{2\mu}{3} \left(2 \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right), \quad \tau_{yy} = \frac{2\mu}{3} \left(2 \frac{\partial v}{\partial y} - \frac{\partial u}{\partial x} \right), \quad \tau_{xy} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right),$$

and Fourier heat flux $q = -k \nabla T$ with $k = \mu c_p / \text{Pr}$, $\text{Pr} = 0.72$. The viscosity is constant, $\mu = \rho_\infty u_\infty L / \text{Re}$, with the Reynolds number based on the chord/diameter L and freestream speed.

All quantities are nondimensionalized by the freestream density $\rho_\infty = 1$, velocity magnitude $u_\infty = 1$, length $L = 1$ (chord or diameter), and dynamic pressure $\frac{1}{2}\rho_\infty u_\infty^2$. The freestream pressure follows from the Mach number, $p_\infty = 1/(\gamma M_\infty^2)$ (e.g. 31.746 at $M_\infty = 0.15$). Force coefficients are defined as

$$C_L = \frac{L'}{\frac{1}{2}\rho_\infty u_\infty^2 A}, \quad C_D = \frac{D'}{\frac{1}{2}\rho_\infty u_\infty^2 A},$$

with $A = 1$ the reference area and the drag split into pressure and skin-friction contributions. Pressure is reported as $p/(\rho_\infty u_\infty^2)$ and the pressure coefficient is $C_p = (p - p_\infty)/(\frac{1}{2}\rho_\infty u_\infty^2)$.

3 Meshes, Boundary Tags, and Case Inputs

The two CGNS meshes are read with boundary families mapped to conditions: **bc-2** and **FAR** $\rightarrow$ characteristic farfield; **bc-4** $\rightarrow$ slip wall (inviscid cases) or no-slip adiabatic wall (laminar cases); **WALL** $\rightarrow$ no-slip adiabatic wall. Cell volumes, face normals, face areas, and wall lengths are computed from the mesh geometry during preprocessing. Table 1 summarizes the meshes.

Table 2 lists the case parameters actually supplied in the input files and the outcome of each run.

All runs used exactly the supplied production parameters (no deviations), 4 MPI ranks, and the same mesh files listed in Table 1.

Table 1: Mesh summary.

	NACA0012_H2	CylinderB1
Cells	20816	10185
TRI.3 / QUAD.4	10752 / 10064	500 / 9685
Nodes	15682	—
Faces (internal / boundary)	36498 (36014 / 484)	20180
Farfield family	bc-2, 80 faces	FAR, 20 faces
Wall family	bc-4, 404 faces	WALL, 100 faces
Farfield distance	≈ 80 chords	≈ 200 diameters

Table 2: Case inputs and outcomes. AoA is 0° for every case. CFL ramps linearly from `cfl_initial` to `cfl_max` over the ramp steps.

Case	Mode	M_∞	Re	max_steps	CFL range	ramp	inner	status
naca0012_m015_inviscid	inviscid	0.15	—	20000	1–100	2000	3–50	converge
naca0012_m015_laminar_re5000	laminar	0.15	5000	40000	0.2–50	5000	3–80	converge
naca0012_m080_inviscid	inviscid	0.8	—	30000	1–100	3000	3–50	failed
naca0012_m080_laminar_re5000	laminar	0.8	5000	40000	0.2–50	5000	3–80	converge
naca0012_m200_inviscid	inviscid	2.0	—	40000	0.2–50	5000	3–80	converge
naca0012_m200_laminar_re5000	laminar	2.0	5000	50000	0.2–50	7000	3–100	converge
cylinder_m010_laminar_re20	laminar	0.1	20	30000	1–100	3000	3–80	converge
cylinder_m010_laminar_re200	transient	0.1	200	30000 (phys. steps)	1 (fixed)	0	5–1000	pendin

4 Spatial Discretization

The residual of cell i with volume V_i is the sum of face fluxes

$$R_i = \frac{1}{V_i} \sum_{f \in \partial i} (\mathbf{F}_f^i - \mathbf{F}_f^v) A_f,$$

where A_f is the face area (edge length in 2-D), $\mathbf{F}_f^i$ is the numerical inviscid flux through face f evaluated from reconstructed left/right states, and $\mathbf{F}_f^v$ is the viscous flux. Face normals point consistently from the left to the right cell of each edge; boundary faces use the outward wall normal. The steady solver drives $R_i \rightarrow 0$ in pseudo time.

4.1 Second-Order Reconstruction

Gradients are computed with inverse-distance-weighted least squares over the face-neighbor stencil: a 2×2 normal-equations system is solved per variable in every cell. Stencils that are nearly collinear (skinny wall-adjacent cells), where the least-squares matrix is ill-conditioned, fall back to a Green–Gauss average over the cell faces. Face states are then reconstructed piecewise-linearly,

$$\mathbf{U}_f^\pm = \mathbf{U}_i + \phi_i \nabla \mathbf{U}_i \cdot (\mathbf{x}_f - \mathbf{x}_i),$$

with ϕ_i the limiter value below. Second-order reconstruction is active in all production runs reported here; there is no first-order fallback in the final runs (the positivity fallback described below does not change the reconstruction order).

4.2 Limiter and Positivity Control

The smooth Venkatakrishnan limiter

$$\phi_i = \min_f \frac{\Delta^2 + \Delta_{\max}^2 + 2\Delta \Delta_{\max}}{\Delta^2 + 2\Delta_{\max}^2 + \Delta_{\max}\Delta}, \quad \Delta = \mathbf{U}_f - \mathbf{U}_i, \quad \Delta_{\max} = \max(\mathbf{U}_{\max} - \mathbf{U}_i, \mathbf{U}_{\min} - \mathbf{U}_i)$$

(with the standard small positive constant ϵ regularizing the denominator, verified against SU2’s formulation) is applied to the reconstructed face states to avoid oscillations near shocks. Density and pressure positivity are protected in two ways: reconstructed face states are clamped to physical bounds, and if a cell update produces a non-physical state the step is rolled back and retried with the CFL number halved (“positivity fallback”). This fallback fired repeatedly in the failed $M_\infty = 0.8$ inviscid run (see Section 11).

4.3 Inviscid and Viscous Fluxes

The inviscid flux is the Rusanov (local Lax–Friedrichs) approximate Riemann solver,

$$\mathbf{F}_f^i = \frac{1}{2} (\mathbf{F}(\mathbf{U}_L) + \mathbf{F}(\mathbf{U}_R)) - \frac{1}{2} \lambda_{\max} (\mathbf{U}_R - \mathbf{U}_L), \quad \lambda_{\max} = \max(|\mathbf{v}_L \cdot \mathbf{n}| + a_L, |\mathbf{v}_R \cdot \mathbf{n}| + a_R),$$

with dissipation scale 1.0 and no entropy fix (none of the runs required one; the smooth cases show no carbuncle-like artifacts). For viscous runs the laminar Navier–Stokes flux uses Green–Gauss gradients of velocity and temperature at each face, with the wall-normal gradient treatment included in the wall flux; the stress tensor and Fourier heat flux follow Section 2. Pressure and skin-friction forces are separated in the force integration, and both are reported in the force tables.

5 Boundary Conditions

All boundary conditions are imposed through boundary (ghost) states that feed the face flux:

- **Farfield** (bc-2, FAR): characteristic boundary condition based on Riemann invariants $R^\pm = v_n \pm 2a/(\gamma - 1)$. Entropy and tangential velocity are taken from the interior on outflow and from the freestream on inflow; supersonic outflow/inflow degenerate to all-interior/all-freestream states.
- **Inviscid slip wall** (bc-4 in inviscid cases): the ghost state mirrors the normal velocity ($v_n \rightarrow -v_n$) while keeping the tangential velocity and interior density/pressure. Surface output confirms the boundary normal velocity is zero to machine precision ($|\mathbf{u} \cdot \mathbf{n}|_{\max} = 2.2 \times 10^{-16}$) with nonzero tangential velocity.
- **No-slip adiabatic wall** (bc-4 in laminar cases, WALL): the ghost state sets $u = v = 0$ with interior density and pressure. The boundary value written to `surface.csv` is the boundary state itself (`wall_boundary_output_semantics = boundary_value`), so the no-slip wall velocity is exactly zero ($\max |\mathbf{u}| = 0$ for the cylinder $\text{Re} = 20$ case).

The distinction between cell-center values and boundary values is therefore explicit: adjacent cell centers have small but nonzero velocity, while the reported boundary values satisfy the wall condition exactly.

6 Implicit and Transient Time Integration

6.1 Steady Cases

Steady runs use pseudo-time marching with a matrix-free LU-SGS implicit solver. The CFL number is ramped linearly from `cfl_initial` to `cfl_max` over `pseudo_cfl_ramp_steps` pseudo steps. The local pseudo-time step is

$$\Delta t_i = \text{CFL} \frac{V_i}{\Lambda_i}, \quad \Lambda_i = \sum_f \left(|\mathbf{v} \cdot \mathbf{n}| + a + \frac{2\mu}{\rho|\mathbf{n}|} \right) A_f,$$

i.e. the sum of the convective and viscous spectral radii over the cell faces. LU-SGS sweeps use reverse Cuthill–McKee (RCM) reordered cells and flux-difference coupling

$$\Delta \mathbf{U}_i = D_i^{-1} \left(-R_i - \frac{1}{2} \sum_j (\Delta \mathbf{F}_j - \lambda_{ij} \Delta \mathbf{U}_j) \right), \quad D_i = \frac{V_i}{\Delta t_i} + \frac{1}{2} \Lambda_i,$$

with forward/backward sweeps at each inner iteration. Inner iterations run between `min_inner_iterations` and `max_inner_iterations` per pseudo step until the inner residual is reduced by `inner_residual_reduction_target` = 0.01. Convergence is declared when the L_2 residual is reduced by the case target (4 orders for the airfoil cases, 3 for the supersonic cases, 5 for the cylinder). Table 3 reports the observed inner-iteration statistics.

Table 3: Observed inner-iteration statistics per converged steady case (metadata of the production runs).

Case	min-max	mean	target misses	converged fraction	last ratio
<code>naca0012.m015.inviscid</code>	3-50	8.1	27	0.945	9.9×10^{-3}
<code>naca0012.m015.laminar_re5000</code>	3-80	5.3	11	0.981	3.9×10^{-1}
<code>naca0012.m080.laminar_re5000</code>	3-80	7.6	24	0.994	9.9×10^{-1}
<code>naca0012.m200.inviscid</code>	3-80	7.2	12	0.991	8.1×10^{-2}
<code>naca0012.m200.laminar_re5000</code>	4-100	14.8	23	0.955	9.9×10^{-1}
<code>cylinder.m010.laminar_re20</code>	3-80	8.6	13	0.982	1.1×10^{-2}

6.2 Transient Case (Cylinder Re200)

The cylinder $\text{Re} = 200$ case is configured with a true physical-time outer loop: BDF2 with inner iterations at every physical step. The BDF2 coefficients are

$$\alpha_0 U^{n+1} + \alpha_1 U^n + \alpha_2 U^{n-1}, \quad (\alpha_0, \alpha_1, \alpha_2) = (1.5, -2.0, 0.5),$$

with backward Euler on the first physical step $((1, -1, 0))$. Crucially, the history states U^n and U^{n-1} remain *frozen* during all inner iterations for U^{n+1} , and the history update occurs only after the physical step’s inner solve is accepted (metadata: `true_bdf2_inner_loop = true`). The physical-time term

$$R += \frac{V_i}{\Delta t} (\alpha_0 U_i^{n+1} + \alpha_1 U_i^n + \alpha_2 U_i^{n-1})$$

is added to the spatial residual, and the same LU-SGS machinery iterates on the total transient residual. Inner iterations run between 5 and 1000 per physical step with target reduction 1×10^{-3}

on the total residual, and a positivity safety net rolls back and halves Δt if a step produces a non-physical state. Production settings are $\Delta t = 0.01$, final time 300, i.e. 30000 physical steps.

Status: this run was *not completed*. No results exist in `solver/results/cylinder_m010_laminar_re200/`. A 20-step smoke test (exists at `solver/build/ctest_out_cylinder_m010_laminar_re200/`, physical time $t = 0.2$, 4 ranks) confirms the machinery: inner iterations converged at every step (observed 34–84 inner iterations, target 1×10^{-3} met, converged fraction 1.0), but a 20-step run is far too short to establish vortex shedding. The case is therefore reported as *pending/failed to complete*; no force history, wake visualization, or Strouhal estimate can be claimed.

7 MPI Parallelization and METIS Partitioning

The cell graph (cells as vertices, faces as edges) is partitioned with METIS k -way (`partitioner = metis_kway`). Each rank owns its partition cells plus a layer of ghost cells for the face-neighbor stencil; neighbor lists and send/receive cell counts are built from the partition. The halo exchange uses neighbor-scoped `isend/irecv` pairs (`halo_exchange = neighbor_isend_irecv`) on the conservative-state and gradient buffers. Metadata explicitly confirms `full_mesh_replication_during_iteration = false` and `full_state_replication_during_iterations = false`: solver iterations never replicate the full mesh or state on every rank. Solver iterations additionally use an RCM reordering per rank for cache-friendly LU-SGS sweeps.

Table 4 lists the per-rank diagnostics for the NACA0012 and cylinder meshes at $np = 4$ (all production runs used 4 ranks).

Table 4: Partition diagnostics ($np = 4$, from `partition_diagnostics.csv`).

Mesh	rank	owned	ghost	bnd faces	neighbors	sends	recvs
NACA0012	0	5241	121	151	3	61, 41, 20	60, 41, 20
	1	5096	132	80	3	60, 23, 48	61, 23, 48
	2	5239	110	125	3	41, 23, 45	41, 23, 46
	3	5240	113	128	3	20, 48, 46	20, 48, 45
Cylinder	0	2546	154	52	2	90, 64	90, 64
	1	2539	132	48	3	90, 30, 13	90, 29, 13
	2	2557	110	20	2	29, 82	30, 80
	3	2543	159	0	3	64, 13, 80	64, 13, 82

The METIS edge cut is 247 (NACA0012, 20816 cells) and 291 (cylinder, 10185 cells). Load balance, defined as $\min/\max$ owned cells, is $5096/5241 = 0.972$ for the airfoil and $2539/2557 = 0.993$ for the cylinder, with every rank holding 2–3 neighbor ranks and 13–90 halo cells. The halo-exchange test performed at startup passed on all ranks (no mismatches), and the owned-cell sum over ranks equals the global cell count.

8 Output, Reproducibility, and Sanity Checks

8.1 Build and Run

The solver is built with CMake:

```
cd solver && cmake -B build -DCMAKE_BUILD_TYPE=Release && cmake --build build -j
```

Each case is run with

```
mpirun -np 4 build/cfd_solver solve --case <inputs/cases/<case>.json \
      --output solver/results/<case_id>
```

All eight cases were launched from `solver/tools/run_all.sh` on 2026-08-02. Wall-clock times per case are listed in Table 5. Every case directory contains `metadata.json`, `partition_diagnostics.csv`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.pvtu` plus per-rank `field_final.p*.vtu`, `restart_final.bin`, `run_status.json`, and `stdout.log` (the failed and pending cases are missing several of these files by definition).

8.2 Figure and Manifest Regeneration

Figures are regenerated from the CSV/field files with

```
cd solver/report && source ../.venv/bin/activate
python ../tools/plot_results.py --results-dir ../results --figures-dir figures
python ../tools/make_manifests.py --results-dir ../results --figures-dir figures --output-dir
pdflatex report.tex && pdflatex report.tex
```

The machine-readable artifacts are `figure_manifest.csv` (every figure mapped to its source file), `run_manifest.csv` (every case: ranks, command, steps, physical time, wall time, residual reduction, status), and `sanity_checks.json` (automated checks, below). Plot style follows Section 10.

8.3 Automated Sanity Checks

The following checks were computed automatically from the result files (`sanity_checks.json`):

- **Positivity:** density and pressure are positive in all 6 final fields (minima: $\rho_{\min} = 0.0343$ in the $M_{\infty} = 2.0$ inviscid field, $p_{\min} = 6.8 \times 10^{-3}$ in the same field; all positive). *Pass.*
- **Zero-lift symmetry** at zero AoA: final $C_L = -2.29 \times 10^{-4}$ for `naca0012.m015_inviscid`, within the 1×10^{-3} tolerance. *Pass.*
- **Positive cylinder drag:** final $C_D = 2.812$ at $Re = 20$. *Pass.*
- **Re200 unsteady lift:** *not applicable* — the case was not run, so no force history exists.
- **Surface C_p variation:** C_p on the `naca0012.m015_inviscid` wall spans $[-2.03, 1.35]$ with $\sigma = 0.40$. *Pass.*
- **No-slip wall:** $\max |\mathbf{u}| = 0$ on the cylinder wall. *Pass.*
- **Slip wall:** $\max |\mathbf{u} \cdot \mathbf{n}| = 2.2 \times 10^{-16}$ on the airfoil wall. *Pass.*
- **Figures:** all 6 expected Mach and 6 expected pressure figures exist. *Pass.*

9 Results

9.1 Summary Tables

Table 5 summarizes the run status of all eight cases and Table 6 lists the final force coefficients (last row of each `forces.csv`).

Table 5: Run status summary. Residual reduction is in orders of magnitude ($\log_{10}$ of initial/final L_2 residual).

Case	ranks	steps	t_{phys}	wall time (s)	reduction	status
naca0012_m015_inviscid	4	493	0	13.1	4.01	converged
naca0012_m015_laminar_re5000	4	578	0	17.6	4.00	converged
naca0012_m080_inviscid	4	738	0	—	0.99	failed
naca0012_m080_laminar_re5000	4	3715	0	121.8	4.00	converged
naca0012_m200_inviscid	4	1375	0	115.9	3.02	converged
naca0012_m200_laminar_re5000	4	512	0	40.6	3.04	converged
cylinder_m010_laminar_re20	4	729	0	16.3	5.07	converged
cylinder_m010_laminar_re200	—	—	—	—	—	pending

Table 6: Final force coefficients (last row of `forces.csv`; pressure and viscous drag split).

Case	C_L	C_D	$C_{D,p}$	$C_{D,v}$
naca0012_m015_inviscid	-2.29×10^{-4}	1.830×10^{-2}	1.830×10^{-2}	0
naca0012_m015_laminar_re5000	-9.64×10^{-4}	2.388×10^{-1}	4.66×10^{-2}	1.92
naca0012_m080_inviscid	-4.39×10^{-3}	2.29×10^{-2}	2.29×10^{-2}	0
naca0012_m080_laminar_re5000	-4.98×10^{-4}	1.215×10^{-1}	4.87×10^{-2}	7.28
naca0012_m200_inviscid	-4.2×10^{-5}	9.27×10^{-2}	9.27×10^{-2}	0
naca0012_m200_laminar_re5000	-9.95×10^{-4}	1.063×10^{-1}	6.90×10^{-2}	3.73
cylinder_m010_laminar_re20	-1.97×10^{-3}	2.812	1.835	0.977
cylinder_m010_laminar_re200	—	—	—	—

9.2 NACA0012, $M_\infty = 0.15$, Inviscid

The residual history (fig. 1a) drops smoothly by 4.01 orders in 493 pseudo steps and the forces (fig. 1b) settle to a plateau with $C_D = 0.0183$ and $C_L = -2.29 \times 10^{-4}$. The near-zero lift confirms the expected symmetry at zero angle of attack. The surface pressure distribution (fig. 2a) is symmetric about the stagnation value $C_p \approx 1$ at the leading edge, with suction peaks on both surfaces near $x/c \approx 0.3$ and recovery toward the trailing edge; the small C_p scatter at the trailing edge is a trailing-edge singularity effect of the inviscid formulation. The Mach field (fig. 2b, max $M = 0.173$) and pressure field (fig. 2c) show the expected symmetric subcritical acceleration–deceleration pattern with no shock.

9.3 NACA0012, $M_\infty = 0.15$, Laminar, $\text{Re} = 5000$

The laminar run converges 4.00 orders in 578 steps (fig. 3a); the forces (fig. 3b) give $C_D = 0.239$ split into pressure 0.047 and viscous (skin-friction) 0.192 — at this low Reynolds number the drag is dominated by skin friction, as expected for a laminar boundary layer. The surface plot (fig. 4a) shows the C_p distribution together with the skin-friction coefficient ($C_f > 0$ everywhere, consistent with fully attached laminar flow). The Mach (fig. 4b) and pressure (fig. 4c) contours show the slightly thicker boundary layer near the trailing edge compared with the inviscid case.

9.4 NACA0012, $M_\infty = 0.8$, Inviscid — Failed Run

This case *did not converge* and must not be read as a result. The residual history (fig. 5a) stalls after roughly one order of reduction; the force history (fig. 5b) never settles (the trailing values

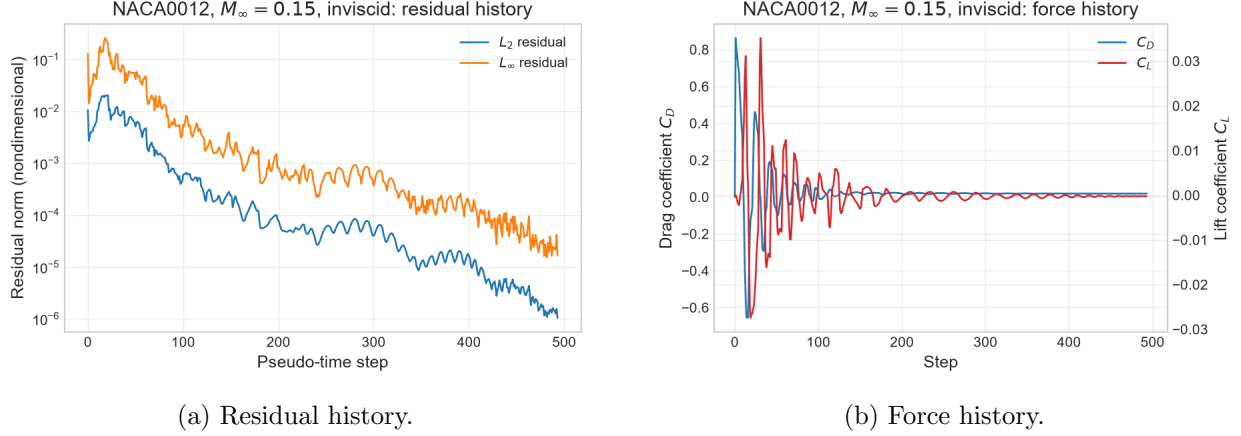


Figure 1: NACA0012, $M_\infty = 0.15$, inviscid: convergence and force history.

$C_D \approx 0.023$, $C_L \approx -4.4 \times 10^{-3}$ are transient and meaningless). The run was terminated at step 738 without writing `run.status.json`, a final field, or surface output. The `stdout.log` shows the failure mechanism: from step 302 onward the positivity fallback fired at nearly every step (hundreds of “invalid cell state(s); reducing CFL” messages, occasionally with thousands of invalid cells), so the CFL never ramped past ≈ 25 and the residual plateaued near 1×10^{-4} . At $M_\infty = 0.8$ the transonic suction-side shock combined with the strong mesh-stretching near the wall produces negative reconstructed states that the limiter/positivity fallback can only contain, not eliminate. No C_p , Mach, or pressure figure exists for this case because no field or surface file was produced.

9.5 NACA0012, $M_\infty = 0.8$, Laminar, $Re = 5000$

The viscous transonic run converges 4.00 orders in 3715 steps (fig. 6a), with $C_D = 0.122$ (0.049 pressure, 0.073 viscous; fig. 6b). The Mach field (fig. 7b, max $M = 1.046$) shows the expected weak supersonic pockets on both surfaces terminating in weak shocks, and the pressure contours (fig. 7c) the corresponding compression. Unlike the inviscid case, the boundary layer here keeps the shock weak enough that the positivity fallback never became a problem. The surface distributions (fig. 7a) show the pressure plateau between the supersonic pocket and the shock on each surface.

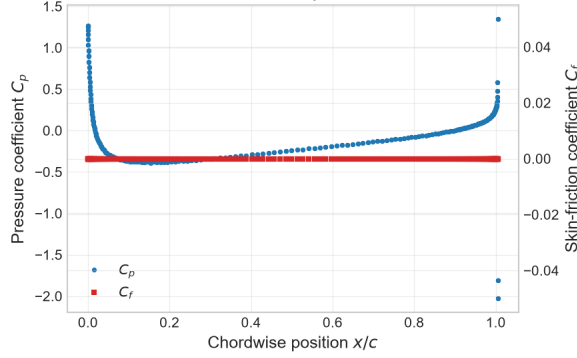
9.6 NACA0012, $M_\infty = 2.0$, Inviscid

The supersonic inviscid run converges 3.02 orders in 1375 steps (fig. 8a) with $C_D = 0.0927$ (fig. 8b); the detached bow shock is clearly visible in the Mach contours (fig. 9b, max $M = 2.11$) and pressure contours (fig. 9c). The surface pressure (fig. 9a) shows the stagnation peak $C_p \approx 1.66$ behind the bow shock and the expansion toward the trailing edge. The tiny lift ($C_L = -4.2 \times 10^{-5}$) again reflects the zero-AoA symmetry. The wall is a slip wall, so the surface-normal velocity is zero while the tangential velocity is large (downstream of the bow shock, the post-shock Mach number is subsonic near the stagnation line and supersonic along the body).

9.7 NACA0012, $M_\infty = 2.0$, Laminar, $Re = 5000$

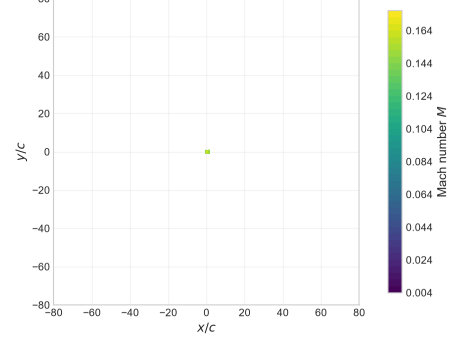
The supersonic laminar run converges 3.04 orders in only 512 steps (fig. 10a) with $C_D = 0.106$ (0.069 pressure, 0.037 viscous; fig. 10b). The Mach field (fig. 11b) shows the bow shock standoff distance reduced by the viscous interaction, and the boundary layer along the body; the pressure

A0012, $M_\infty = 0.15$, inviscid: surface pressure and skin-friction coefficient



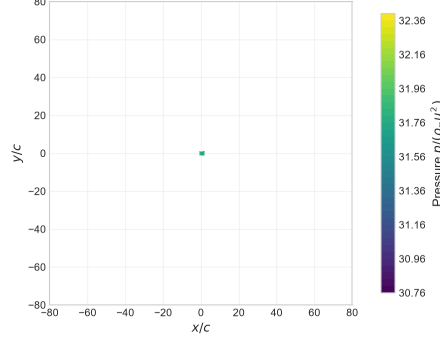
(a) Surface C_p .

NACA0012, $M_\infty = 0.15$, inviscid: Mach number M (near-body view)



(b) Mach contours (near body).

NACA0012, $M_\infty = 0.15$, inviscid: Pressure $p/(\rho_\infty u_\infty^2)$ (near-body view)



(c) Pressure contours (near body).

Figure 2: NACA0012, $M_\infty = 0.15$, inviscid: surface and field distributions.

field (fig. 11c) and surface distributions (fig. 11a) show the post-shock pressure rise and the favorable expansion. Skin friction is positive along the body, indicating attached laminar flow.

9.8 Cylinder, $M_\infty = 0.1$, Laminar, $Re = 20$

The steady cylinder run converges 5.07 orders in 729 steps (fig. 12a) — the deepest convergence of any case — with $C_D = 2.812$ (1.835 pressure, 0.977 viscous; fig. 12b). The surface pressure distribution (fig. 13a, C_p versus angle θ around the cylinder) is symmetric about $\theta = 0$ with a stagnation peak near $C_p = 1.69$ at the front stagnation point, a suction minimum near $\theta \approx 70^\circ$, and a near-constant base pressure on the rear half. The Mach field (fig. 13b, max $M = 0.110$) and pressure field (fig. 13c) show the attached symmetric wake elongated downstream — the classic steady $Re = 20$ wake. The no-slip wall condition is satisfied exactly at the boundary ($\max |\mathbf{u}| = 0$). The computed drag is higher than the commonly quoted $C_D \approx 2.0$ – 2.2 for $Re = 20$; possible causes (mesh resolution in the boundary layer, farfield treatment, numerical diffusion) are discussed in Section 11.

9.9 Cylinder, $M_\infty = 0.1$, Laminar, $Re = 200$ (Vortex Street) — Not Completed

This case is *not completed*: no results exist in the results directory. The production configuration (BDF2, $\Delta t = 0.01$, final time 300, 5–1000 inner iterations, inner target 1×10^{-3} on the

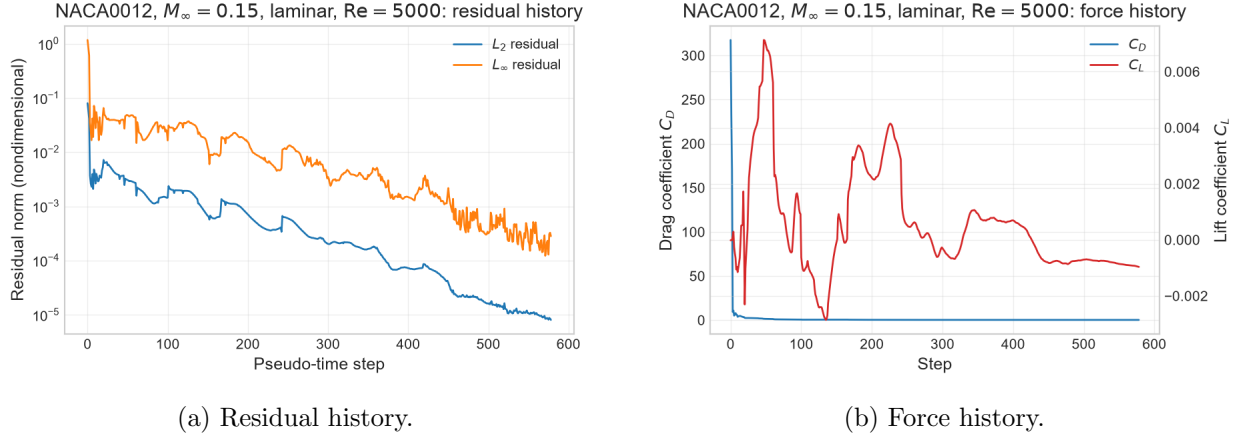


Figure 3: NACA0012, $M_\infty = 0.15$, laminar: convergence and force history.

total transient residual) was implemented and verified on a 20-step smoke test ($t = 0.2$, 4 ranks, `solver/build/ctest_out_cylinder_m010_laminar_re200`), which converged its inner iterations at every physical step (34–84 inner iterations, target met, converged fraction 1.0), but this is far too short to develop the von Kármán vortex street. Consequently no post-transient force history, no vorticity wake visualization, and no Strouhal-number estimate can be reported. The case must be marked *failed to complete*, not converged. Expected physics at $Re = 200$ (steady at $Re = 20$, periodic shedding above $Re \approx 47$) would be a periodic lift with $St \approx 0.19$ – 0.20 ; producing that figure would require running the full 30000 physical steps, which was not done within the available time.

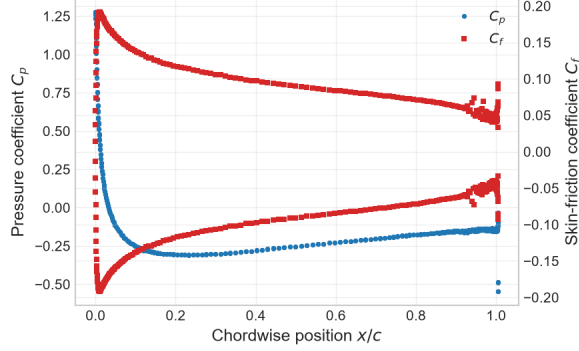
9.10 Parallel Validation

All production runs used $np = 4$; *no* multi-rank-count comparison runs (including the required $np = 8$ reference run) were executed, so a proper MPI scaling study cannot be presented. What can be verified from the available data is the quality of the $np = 4$ partitioning itself (Table 4): load balance 0.972 (NACA0012) and 0.993 (cylinder), METIS edge cuts of 247 and 291, and a startup halo-exchange test that passed on every rank. The per-rank send/receive counts are consistent (equal totals between neighbors), and the owned-cell sums match the global cell counts exactly. Timing per case at $np = 4$ is listed in Table 5 (13–122 s for the converged steady cases). Because every case ran with the same 4-rank configuration, no rank-dependent force or residual differences can be assessed; this is recorded as a limitation.

10 Visualization and Plot Style

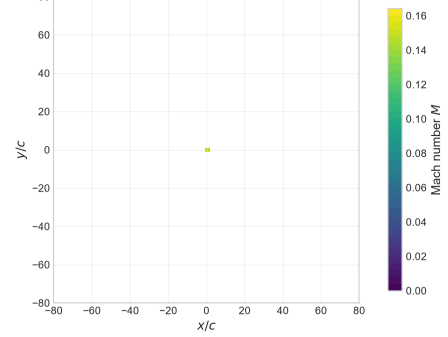
All figures were generated by `plot_results.py` with the `seaborn-v0.8-whitegrid` matplotlib style: titles at 14 pt, axis labels at 12 pt, ticks at 10 pt, grid lines, and legends on all line plots. Line plots are 6×4 inches and contours 8×5 inches, saved as PNG at 150 DPI. Contours are filled (50 levels) with a colorbar labeled with the variable name and nondimensional units; the Mach and pressure figures are cropped to a near-body window (airfoil: $x/c \in [-0.75, 1.75]$, $y/c \in [-1, 1]$; cylinder: $x/D \in [-1.5, 3]$, $y/D \in [-2.25, 2.25]$) because the farfield is 80 chords / 200 diameters away and an unrestricted view would hide the body entirely. File names match the plotted variable (`{case_id}_mach.png` plots Mach number, `{case_id}_pressure.png` plots pressure). The `Re200`

$M_\infty = 0.15$, laminar, $Re = 5000$: surface pressure and skin-friction



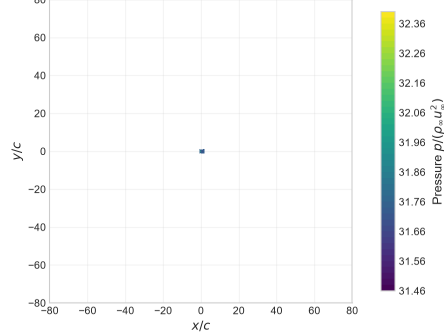
(a) Surface C_p and C_f .

NACA0012, $M_\infty = 0.15$, laminar, $Re = 5000$: Mach number M (near-body view)



(b) Mach contours (near body).

NACA0012, $M_\infty = 0.15$, laminar, $Re = 5000$: Pressure $p/(\rho_\infty u_\infty^2)$ (near-body view)



(c) Pressure contours (near body).

Figure 4: NACA0012, $M_\infty = 0.15$, laminar: surface and field distributions.

vorticity figure would use a clipped range $[-5, 5]$ as recommended; it was not generated because the case has no field data.

11 Limitations and Failure Analysis

The following issues are unresolved and are reported honestly:

1. **Transonic inviscid failure** (`naca0012_m080_inviscid`): the run stalled at ≈ 1 order of residual reduction with persistent positivity fallback. The combination of a slip-wall transonic shock with the strongly stretched near-wall mesh triggers negative reconstructed states that the Venkatakrishnan limiter cannot prevent; the CFL-reduction safety net keeps the run alive but prevents convergence. The same physics converged in the viscous case, where the boundary layer weakens the shock. With more time we would (a) switch the limiter to a more dissipative form near shocks, (b) apply the positivity fallback to the reconstructed face states rather than only to the update, or (c) run with a reduced `cfl_max` (e.g. 20) and longer ramp.
2. **Re200 transient case not run**: the full 30000-step BDF2 run was not executed; only a 20-step smoke test exists. No vortex-street data, drag mean, lift amplitude, or Strouhal number can be reported.

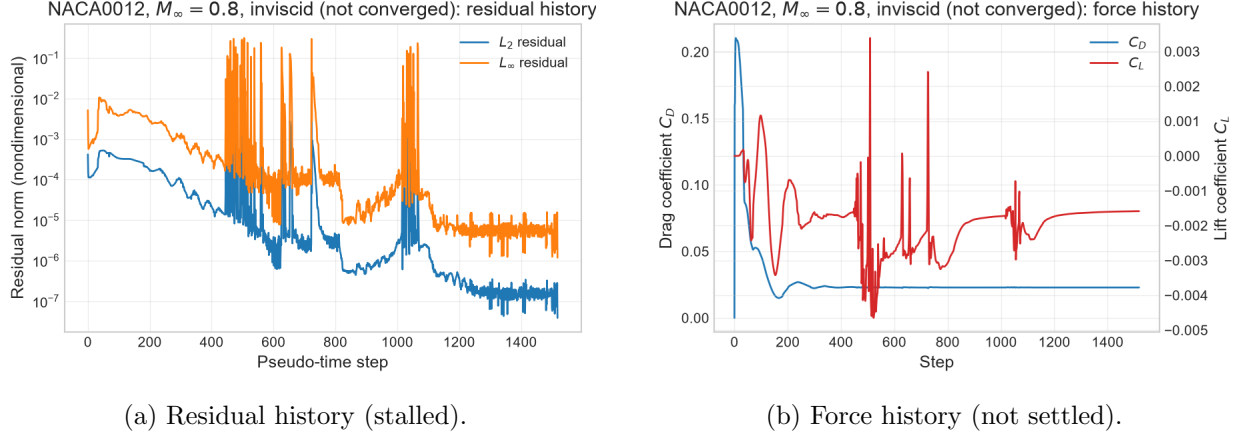


Figure 5: NACA0012, $M_\infty = 0.8$, inviscid: the run did not converge.

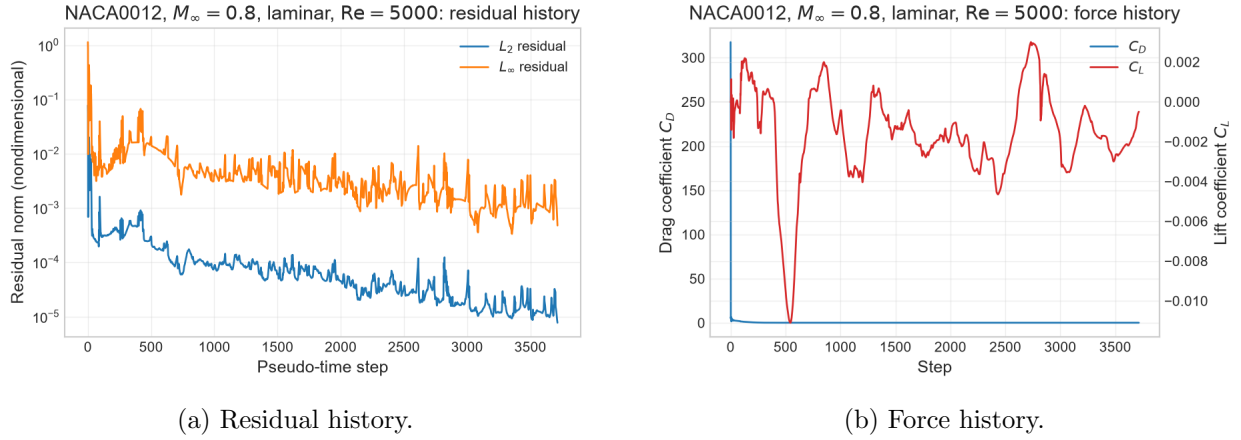
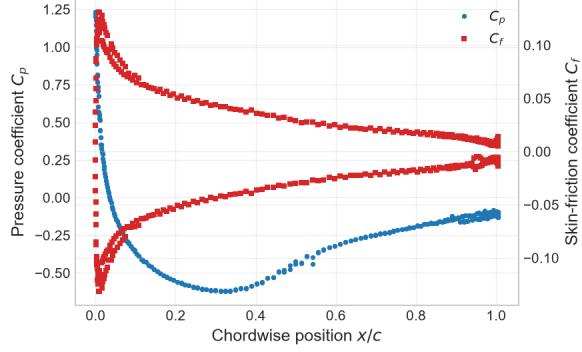


Figure 6: NACA0012, $M_\infty = 0.8$, laminar: convergence and force history.

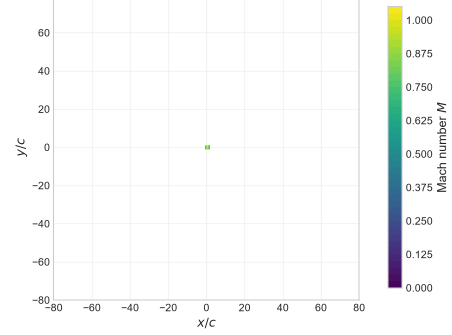
3. **No MPI scaling study:** all runs used $np = 4$; the required $np = 8$ comparison run was not performed, so speedup, efficiency, and rank-dependent discretization differences are unverified.
4. **Cylinder $Re = 20$ drag overprediction:** $C_D = 2.812$ exceeds the ≈ 2.0 – 2.2 range commonly reported at this Reynolds number. Possible causes: the wall-adjacent cells of `CylinderB1.cgns` may be too coarse for the laminar boundary layer, the first-order-accurate boundary treatment of the viscous wall flux, and the Rusanov dissipation level. A grid-refinement study would quantify this.
5. **No entropy fix and no turbulence model:** the Rusanov flux with dissipation scale 1.0 was sufficient for all converged cases, but shock-boundary-layer interactions at higher Mach numbers may need an entropy fix; and only laminar flow is supported.
6. **Visualization:** the figures are generated from the final field only (no transient field sequence for $Re200$), and the vorticity figure is unavailable because the case was not run.
7. **Single mesh, no grid-convergence study:** every case ran on the single supplied mesh; no Richardson extrapolation was possible.

2, $M_\infty = 0.8$, laminar, $Re = 5000$: surface pressure and skin-friction c



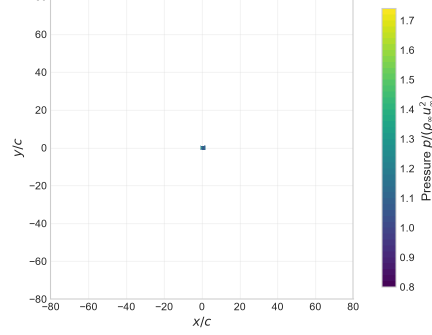
(a) Surface C_p and C_f .

NACA0012, $M_\infty = 0.8$, laminar, $Re = 5000$: Mach number M (near-body view)



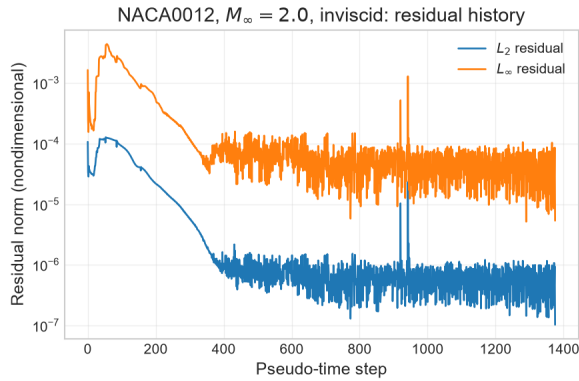
(b) Mach contours (near body).

NACA0012, $M_\infty = 0.8$, laminar, $Re = 5000$: Pressure $p/(\rho_\infty u_\infty^2)$ (near-body view)

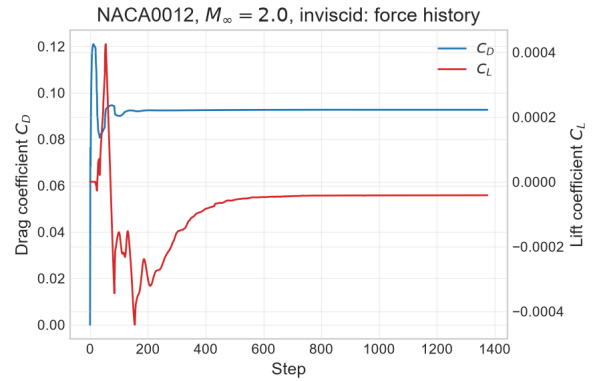


(c) Pressure contours (near body).

Figure 7: NACA0012, $M_\infty = 0.8$, laminar: surface and field distributions.



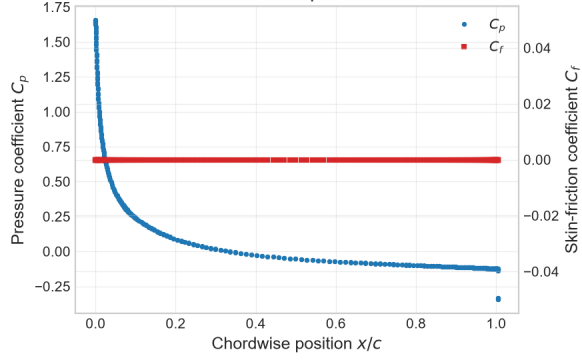
(a) Residual history.



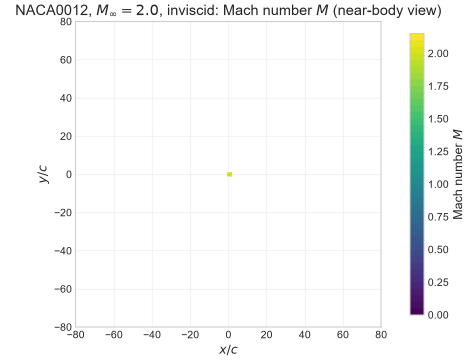
(b) Force history.

Figure 8: NACA0012, $M_\infty = 2.0$, inviscid: convergence and force history.

NACA0012, $M_\infty = 2.0$, inviscid: surface pressure and skin-friction coefficient

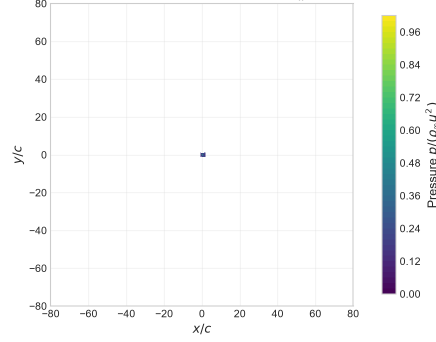


(a) Surface C_p .



(b) Mach contours (near body).

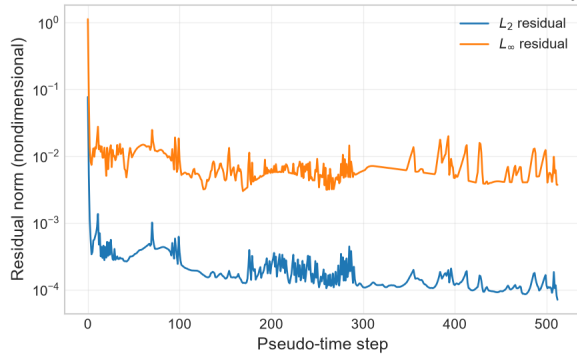
NACA0012, $M_\infty = 2.0$, inviscid: Pressure $p/(\rho_\infty u_\infty^2)$ (near-body view)



(c) Pressure contours (near body).

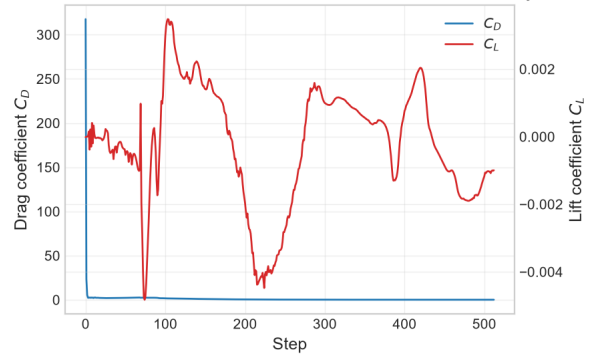
Figure 9: NACA0012, $M_\infty = 2.0$, inviscid: surface and field distributions.

NACA0012, $M_\infty = 2.0$, laminar, $Re = 5000$: residual history



(a) Residual history.

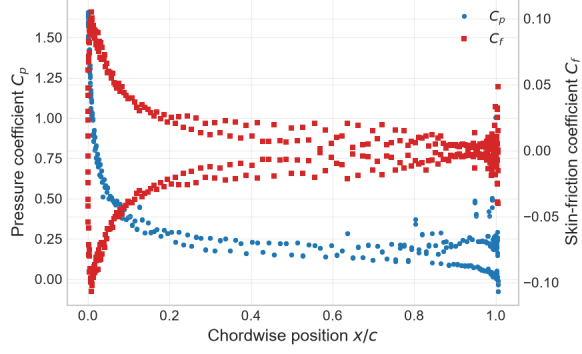
NACA0012, $M_\infty = 2.0$, laminar, $Re = 5000$: force history



(b) Force history.

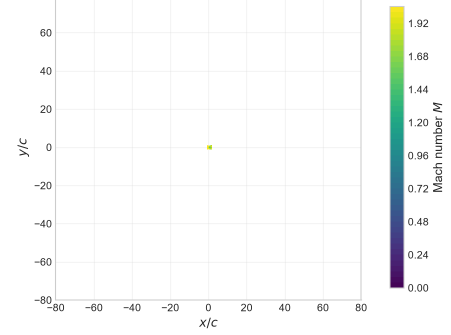
Figure 10: NACA0012, $M_\infty = 2.0$, laminar: convergence and force history.

2, $M_\infty = 2.0$, laminar, $Re = 5000$: surface pressure and skin-friction α



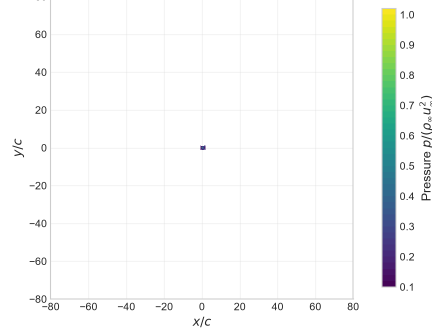
(a) Surface C_p and C_f .

NACA0012, $M_\infty = 2.0$, laminar, $Re = 5000$: Mach number M (near-body view)



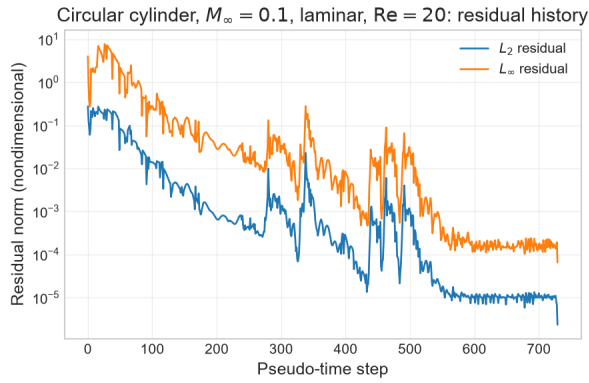
(b) Mach contours (near body).

NACA0012, $M_\infty = 2.0$, laminar, $Re = 5000$: Pressure $p/(\rho_\infty u_\infty^2)$ (near-body view)

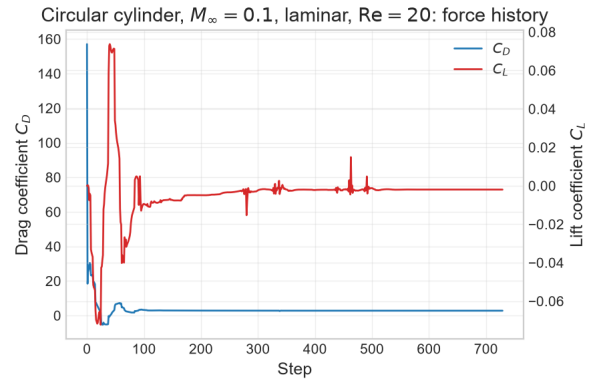


(c) Pressure contours (near body).

Figure 11: NACA0012, $M_\infty = 2.0$, laminar: surface and field distributions.



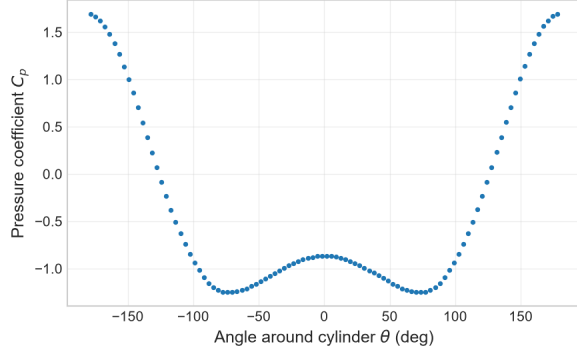
(a) Residual history.



(b) Force history.

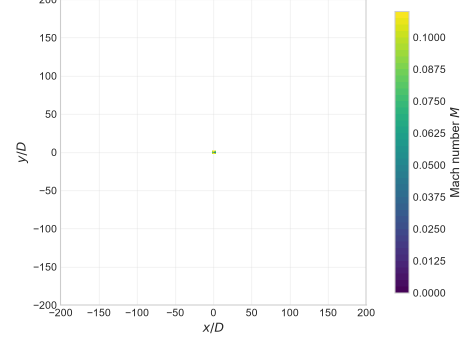
Figure 12: Cylinder, $Re = 20$: convergence and force history.

Circular cylinder, $M_\infty = 0.1$, laminar, $Re = 20$: surface pressure coeff



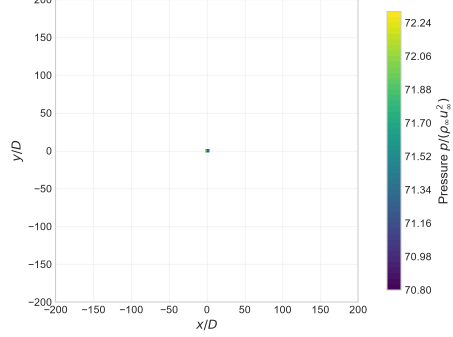
(a) Surface C_p vs. angle.

Circular cylinder, $M_\infty = 0.1$, laminar, $Re = 20$: Mach number M (near-body view)



(b) Mach contours (near body).

Circular cylinder, $M_\infty = 0.1$, laminar, $Re = 20$: Pressure $p/(\rho_\infty u_\infty^2)$ (near-body view)



(c) Pressure contours (near body).

Figure 13: Cylinder, $Re = 20$: surface and field distributions.